

Erdős Problem #14

EP-14

L1 - TRACTABLE

OPEN

Number Theory

ORIGINAL PROBLEM STATEMENT

Let $A \subseteq \mathbb{N}$. Let $B \subseteq \mathbb{N}$ be the set of integers which are representable in exactly one way as the sum of two elements from A .

Is it true that for all $\epsilon > 0$ and large N $|\{1, \dots, N\} \setminus B| \gg_{\epsilon} N^{1/2-\epsilon}$? Is it possible that $|\{1, \dots, N\} \setminus B| = o(N^{1/2})$?

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Archive: <https://www.unsignedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>